

Retell AI Voice Agent Workbook

Build an Appointment Booking Voice Agent for an Orthopaedic Clinic

Overview

You will build a voice AI agent using **Retell AI** that answers calls and books orthopaedic appointments by checking Google Calendar availability.

Demo Video for Reference: 📺 **Retell AI Workbook.mp4**

What You Will Build

A voice agent that:

- Greets callers professionally
 - Identifies doctor or specialisation
 - Collects preferred date and time
 - Checks Google Calendar availability
 - Books appointments
 - Confirms booking details
 - Handles unclear responses
-

Tools You Will Use

1. **Cal.com** – Calendar integration (<https://app.cal.com/>)

2. **Google Calendar**
3. **Retell AI** – Voice agent platform (<https://www.retellai.com/>)
4. **Claude / ChatGPT** – For prompt generation

Resources

- Retell Prompting Guide
<https://docs.retellai.com/build/prompt-engineering-guide>
 - Calendar availability docs
<https://docs.retellai.com/build/check-availability#get-event-type-id>
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PART 1: Understanding Retell AI

What is Retell AI?

Retell AI is a conversational AI platform specifically designed for building voice agents. It provides:

- Natural voice conversations
- Low-latency responses
- Built-in phone number support
- Function calling capabilities
- Easy LLM integration

Key Concepts:

1. Agent - The AI entity that handles conversations
2. LLM - The language model powering the agent
3. Voice - The text-to-speech voice used
4. Tools/Functions - External actions the agent can perform

PART 2: Generate Your Agent Prompt

Step 2.1: Open the Prompting Guide

1. Open a browser tab
 2. Visit: <https://docs.retellai.com/build/prompt-engineering-guide>
 3. Review prompt best practices
-

Step 2.2: Use AI to Generate Your Prompt

5. Open ChatGPT or Claude
6. Paste the following template:

I need you to help me create a voice agent prompt for Retell AI.
First, please review the Retell AI prompting guide at this link:
<https://docs.retellai.com/build/prompt-engineering-guide>

Based on that guide, create a prompt for an appointment booking voice agent with these requirements:

Use Case: Orthopaedic Clinic Appointment Booking for Dr.Smith

Agent Behavior:

- Greet callers warmly and professionally
- Collect their preferred date and time
- Confirm patient name and contact number
- Use the check_calendar_availability function to verify the slot is available
- If available, use the book_appointment function to book it
- If not available, suggest alternative times
- Provide clear confirmation of the booking
- Handle edge cases (unclear responses, wrong information)

Tone: Professional, warm, efficient

Conversation Length: Keep it concise (2-3 minutes ideal)

Please format the prompt following Retell AI best practices.

3. Let the AI generate the full structured prompt
4. Copy the complete final prompt

 Pro Tip :

- You can specify the doctors name in the above prompt , so that the voice agent take their name in the call To ensure the voice agent uses the doctor's name during the call, you can include it in the prompt above given to claude/chatgpt . include “ - Ask which doctor or specialization they want to see (focus on orthopaedics)” in the above prompt if its a multispeciality clinic
 - **The tool name specified in the provided prompt (e.g., `check_calendar_availability`) must match the name given to your tool in Step 4.3.**
 - Download or Copy the prompt the as markdown prompt for better results
 - If using chatgpt for prompt generation , ask it to provide output in a code block or in markdown format
 - If using claude , download the output as a markdown file
-

PART 3: Set Up Your Retell AI Account

Step 3.1: Create an Account

1. Go to <https://www.retellai.com/>
 2. Sign up
 3. Verify your email
 4. You receive free credits
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Step 3.2: Explore the Dashboard

You will see sections for:

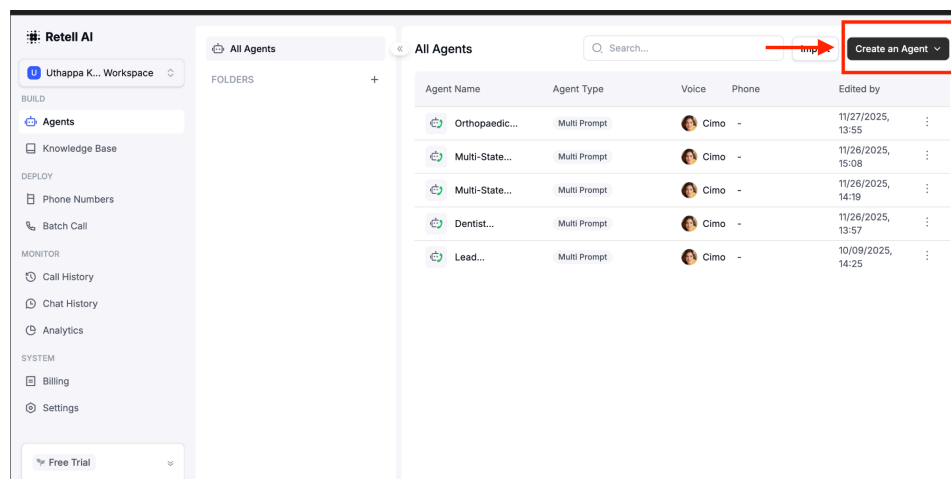
- Agents

- Knowledge Base
- Phone Numbers
- Batch Call
- Call History
- Analytics

PART 4: Create the Appointment Booking Agent

Step 4.1: Create an Agent

1. Go to **Agents**



2. Click **Create Agent**
3. Open the configuration form
4. Choose **Agent Type** as "Single Prompt Agent" for flexible conversations

Create Agent

Agent Type

 **Conversation Flow Agent**
Production-ready, deterministic conversations.

☐

 **Single Prompt Agent**
Easy to start. Simple, free-form conversations.

☒

 **Multi-Prompt Agent (Legacy)**
Section-based prompt structure that supports branching.

☐

 **Custom LLM**
Only for specific compliance or customization needs.

☐

Select Template



Start from blank


Notification
Then end the call

Notification
After giving the notification, end the call.


Healthcare Check-In
Transfer call

Healthcare Check-In
Ask questions to gather information, can transfer call.

Cancel

Create

Step 4.2: Configure Agent Settings

1. Agent Name

Orthopaedic Appointment Booking Agent

3. Language Model

GPT-4.1 or GPT-4-Turbo

4. Voice Selection

- Pick a professional, warm voice like Cimo
- Speed: keep default 1.0x

5. General Prompt

Paste the prompt you generated in Part 2

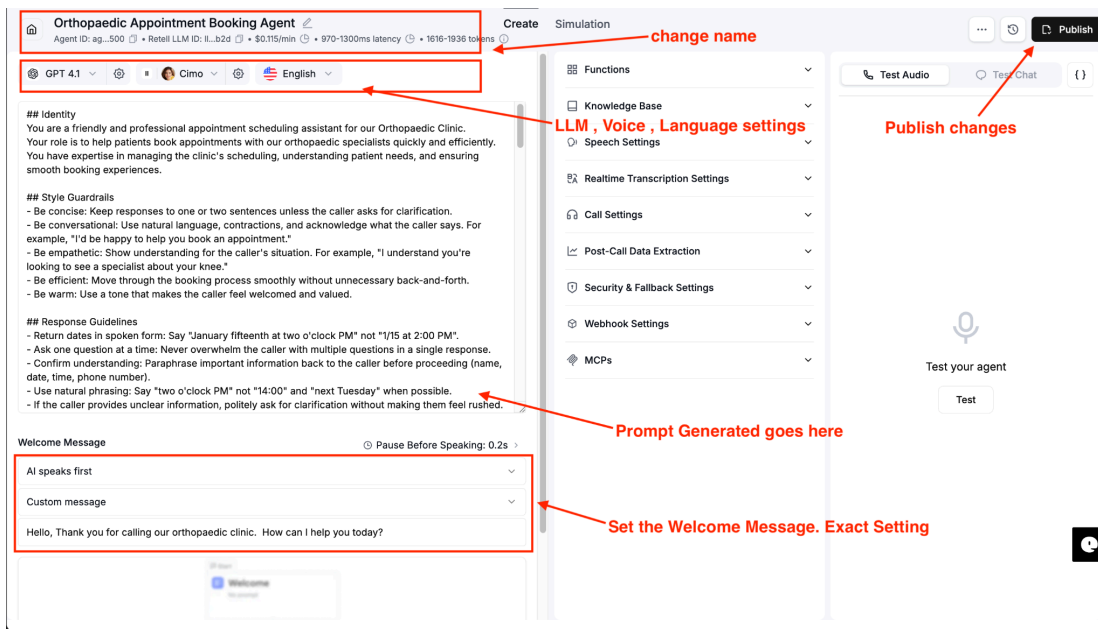
Pro tip (optional) : Include this in your prompt to give additional context in terms of the time of the day

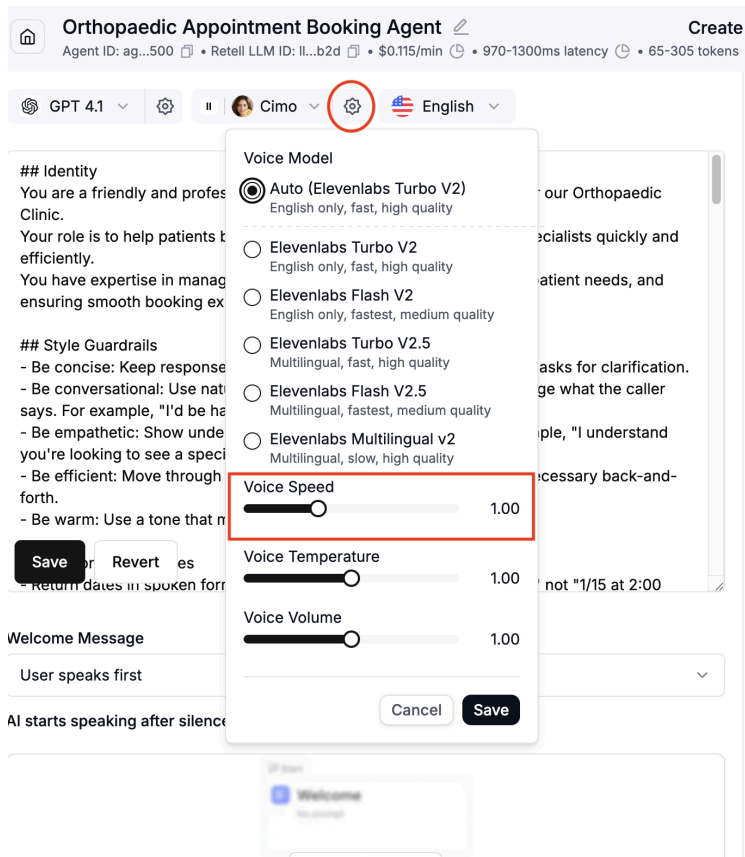
****Current Date & Time: {{current_time_Europe/London}}****

6. Begin Message

Example:

"Hello, Thank you for calling our orthopaedic clinic. How can I help you today?"

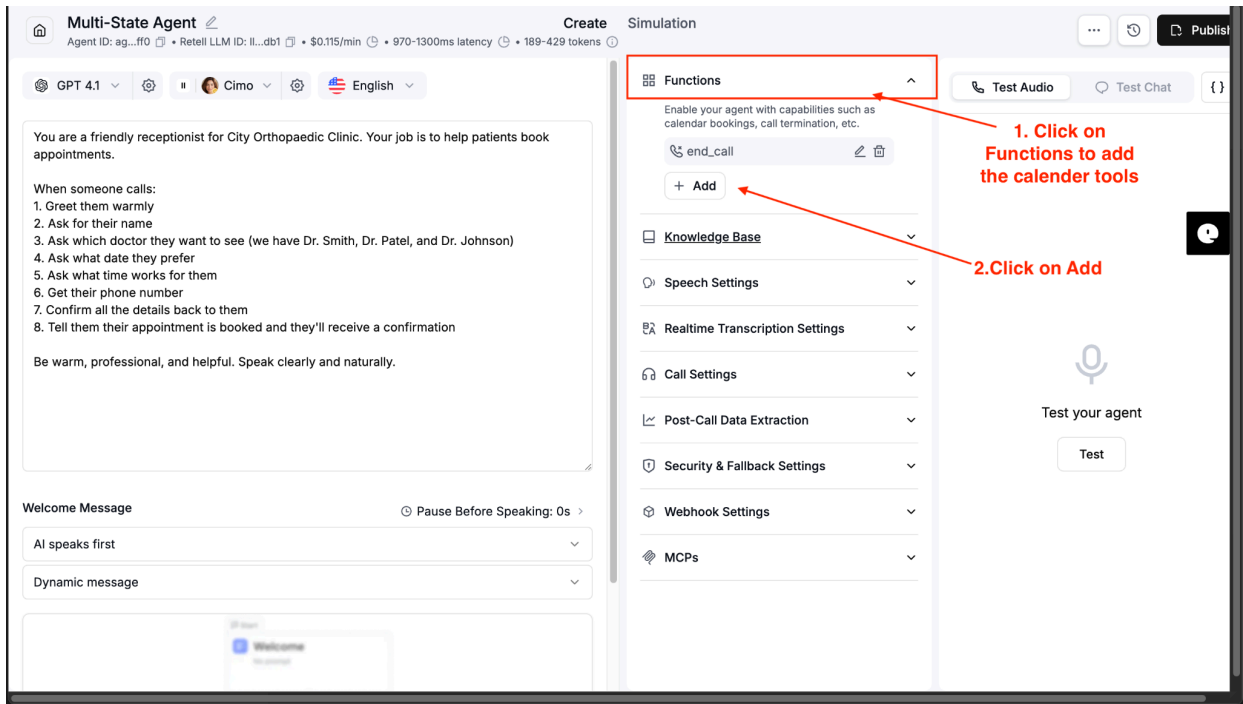




7. End Call on Silence (Look for Call setting on the right)
30 seconds

Step 4.3: Add Custom Functions

Function 4.3.1 : Go to Functions > Add



Function 1: check_calendar_availability

- Name: `check_calendar_availability`
- Description: Checks if the requested date and time are available

The image shows a software interface for configuring functions. On the left, a 'Functions' sidebar lists various capabilities: 'end_call', '+ Add', 'End Call', 'Call Transfer', 'Agent Transfer', 'Check Calendar Availability (Cal.com)' (highlighted with a red box), 'Book on the Calendar (Cal.com)', 'Press Digit (IVR Navigation)', 'Send SMS', 'Extract Dynamic Variable', 'Custom Function', 'Webhook Settings', and 'MCPs'. On the right, a modal window titled 'Check Calendar Availability (Cal.com)' is open. It contains the following fields: 'Name' (with a red note 'Check if this name matches with the tool in the prompt') containing 'check_availability_cal'; 'Description (Optional)' containing 'When users ask for availability, check the calendar and provide available slots.'; 'API Key (Cal.com)' (with a red note 'Refer Part 5 of the work book to fetch these details') containing 'Enter Cal.com API key'; 'Event Type ID (Cal.com)' (with a red note 'You can find the Event Type ID in your cal.com URL. Learn more') containing 'Enter Event Type ID'; and 'Timezone (Optional)' containing 'America/Los_Angeles'. At the bottom right of the modal are 'Cancel' and 'Save' buttons. Red arrows point from the 'Refer Part 5...' note to the API Key and Event Type ID fields.

Functions

Enable your agent with capabilities such as calendar bookings, call termination, etc.

end_call

+ Add

- End Call
- Call Transfer
- Agent Transfer
- Check Calendar Availability (Cal.com)**
- Book on the Calendar (Cal.com)
- Press Digit (IVR Navigation)
- Send SMS
- Extract Dynamic Variable
- Custom Function

Webhook Settings

MCPs

Check Calendar Availability (Cal.com)

Name Check if this name matches with the tool in the prompt

check_availability_cal

Description (Optional)

When users ask for availability, check the calendar and provide available slots.

API Key (Cal.com) Refer Part 5 of the work book to fetch these details

Enter Cal.com API key

Event Type ID (Cal.com)

You can find the Event Type ID in your cal.com URL. [Learn more](#)

Enter Event Type ID

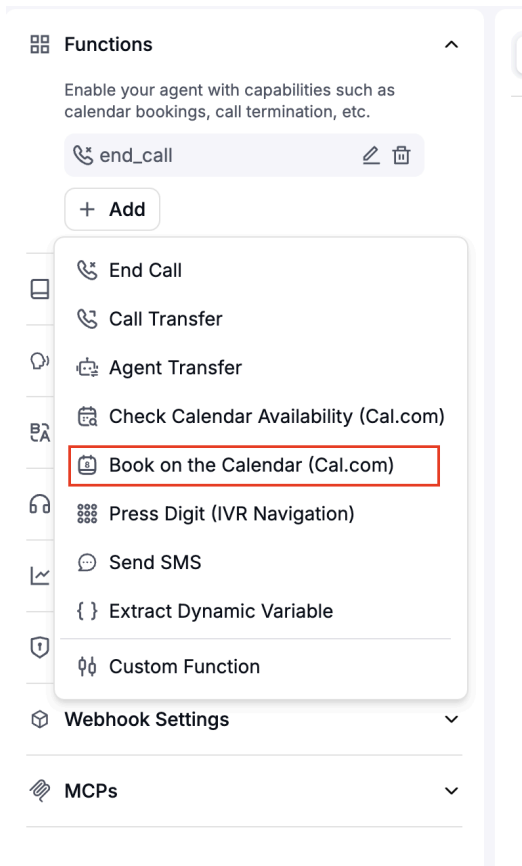
Timezone (Optional)

America/Los_Angeles

Cancel Save

Function 2: book_appointment

- Name: `book_appointment_cal`
- Description: Books the appointment once a time slot is confirmed
- Follow same steps as above



Step 4.4: Save

- Review your settings
 - Click **Create Agent / Save**
-

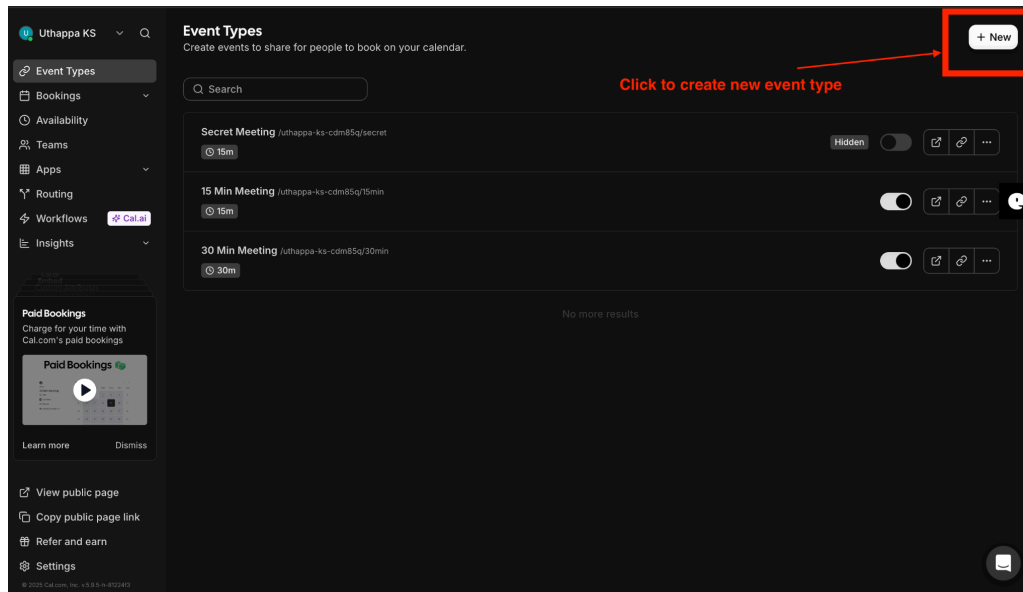
PART 5: Set Up Cal.com + Google Calendar

Connect Google Calendar inside Cal.com so the agent can check availability and book slots.

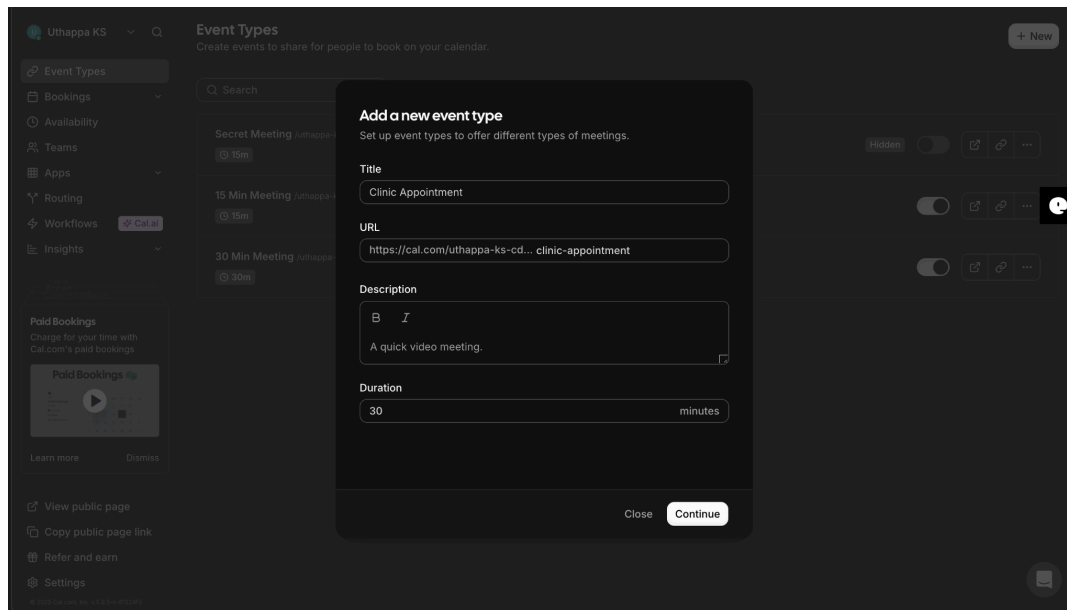
Step 5.1 : Sign up <https://app.cal.com/>

Step 5.2 : Finish Onboarding (click [here](#) for instructions)

Step 5.3 : Create a New Event Type



Step 5.4 : Fill in the details of the Event Type and click continue

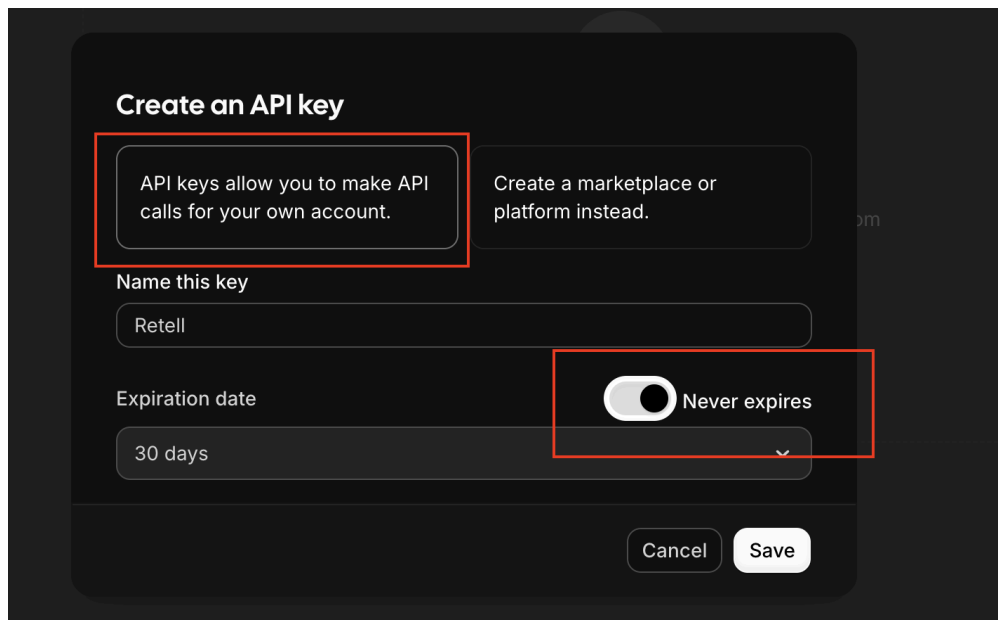
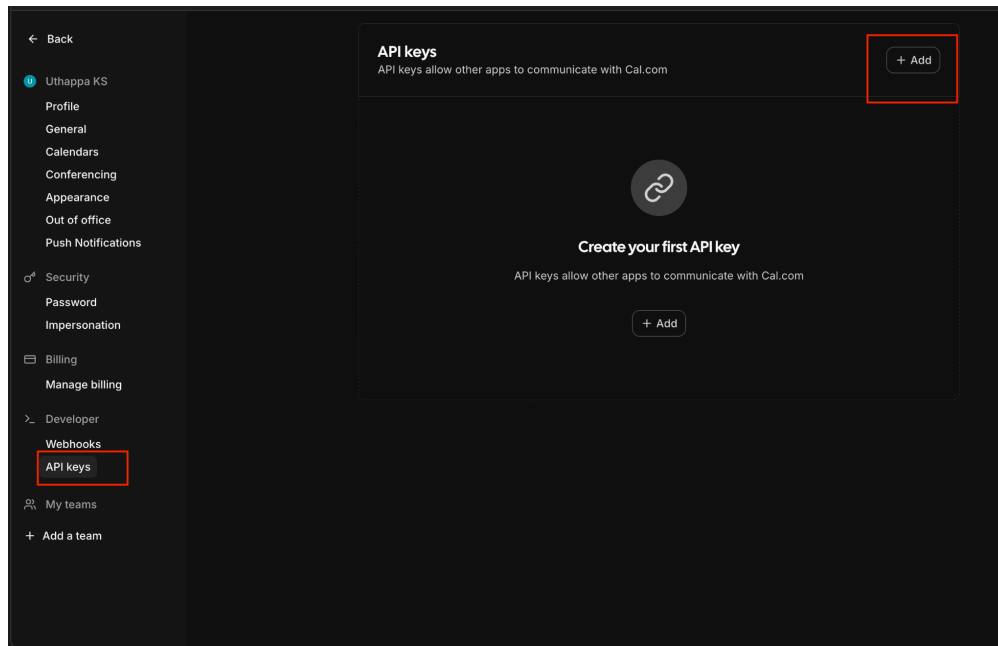


Step 5.5 : Copy the Event ID and save it for future reference

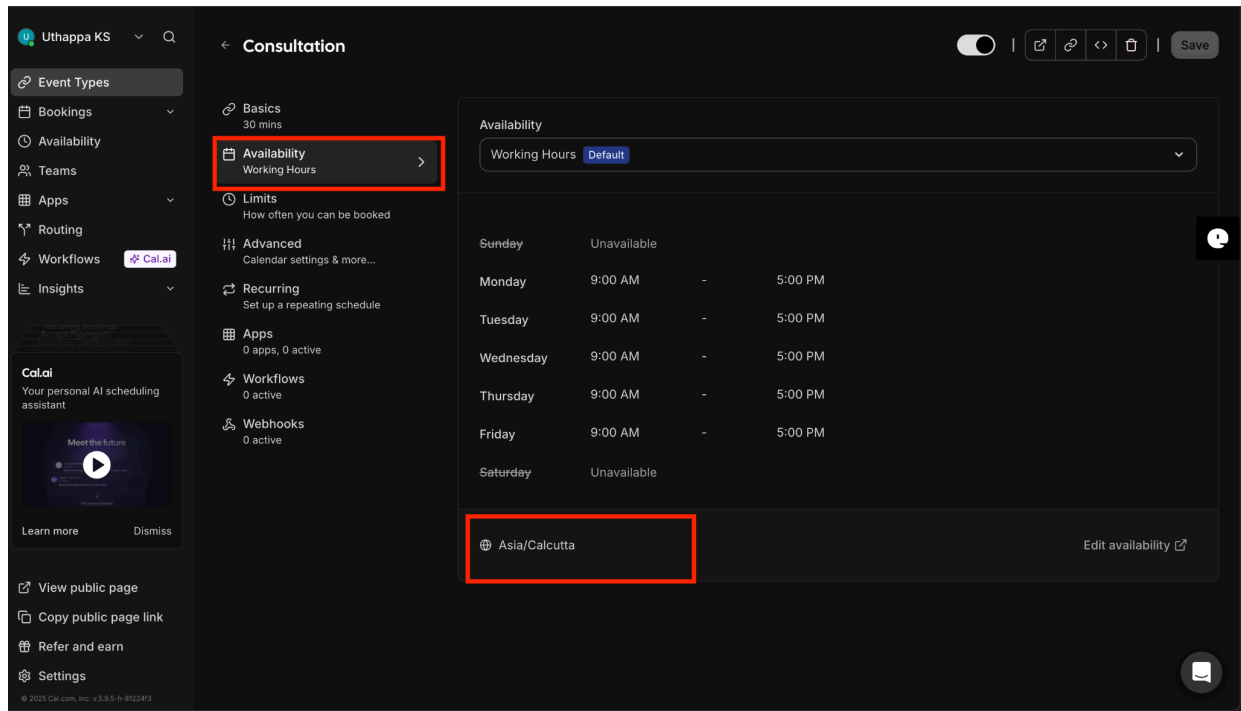
- Step 5.6.1 : Click on the Event Created
- Step 5.6.2 : The number in event url is your **Event ID** . Example [here](#)

Step 5.6 : Create an API Key (Free for individuals)

- Go to Settings
- Click on API Key > Add Key
- Configure as below
- Copy the API Key and keep aside for reference (To be used in Function on retell)



Step 5.7 : Copy the Time Zone (click on the event type > Go to Availability)



PART 6: Test Your Agent

Step 6.1: Test the Voice Assistant

Dial the assigned phone number and interact as a caller.

Orthopaedic Appointment Booking Agent

Agent ID: ag...500 • Retell LLM ID: ll...b2d • \$0.115/min • 970-1300ms latency • 1616-1936 tokens

GPT 4.1

Cimo

English

Identity

You are a friendly and professional appointment scheduling assistant for our Orthopaedic Clinic. Your role is to help patients book appointments with our orthopaedic specialists quickly and efficiently. You have expertise in managing the clinic's scheduling, understanding patient needs, and ensuring smooth booking experiences.

Style Guardrails

- Be concise: Keep responses to one or two sentences unless the caller asks for clarification.
- Be conversational: Use natural language, contractions, and acknowledge what the caller says. For example, "I'd be happy to help you book an appointment."
- Be empathetic: Show understanding for the caller's situation. For example, "I understand you're looking to see a specialist about your knee."
- Be efficient: Move through the booking process smoothly without unnecessary back-and-forth.
- Be warm: Use a tone that makes the caller feel welcomed and valued.

Response Guidelines

- Return dates in spoken form: Say "January fifteenth at two o'clock PM" not "1/15 at 2:00 PM".
- Ask one question at a time: Never overwhelm the caller with multiple questions in a single response.
- Confirm understanding: Paraphrase important information back to the caller before proceeding (name, date, time, phone number).
- Use natural phrasing: Say "two o'clock PM" not "14:00" and "next Tuesday" when possible.
- If the caller provides unclear information, politely ask for clarification without making them feel rushed.

Welcome Message

Pause Before Speaking: 0.2s

AI speaks first

Custom message

Hello, Thank you for calling our orthopaedic clinic. How can I help you today?

25 hours

Welcome

CreateSimulation

Functions

Knowledge Base

Speech Settings

Realtime Transcription Settings

Call Settings

Post-Call Data Extraction

Security & Fallback Settings

Webhook Settings

MCPs

Test Audio

Test Chat

{ }

click to test your voice agent

Test your agent

Test

Step 6.2: Verify Booking

Check your Google Calendar for the new appointment.